Canadian Immigration Statistics — Database Coverage Report

Breakdowns, geography, and temporal grain in `series_facts`, reconciled against the source downloads

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1 Purpose and methodology

This report describes what actually made it into `app/data/immigration.duckdb`'s `series_facts` table - the deduplicated, per-(theme, series) table the Shiny app queries directly - and cross-checks it against the raw source downloads to see whether any breakdown IRCC/IRB actually publishes failed to survive the pipeline.

A quick vocabulary note, since the app and the pipeline scripts use these words in a specific sense:

- **Theme** - a dataset grouping (e.g. Permanent Residents, Study Permit Holders) - the app's top-level "Select Theme" picker.
- **Series** - a corrected, de-duplicated label (`catalog$series_label`, baked in by `data-raw/06_build_label_lookups.R`) that can bundle several raw source tables together when they're genuinely the same underlying series published as separate sheets (e.g. one sheet "by Sex", another "by Immigration Category") - the app's "Series" picker.
- **Breakdown** (a.k.a. **facet**) - a non-geographic dimension a series can be split by (e.g. Gender, Immigration Category) - the app's "Break down by" picker. `series_facts$facet = "Total"` is the no-breakdown grand total, and is excluded from the breakdown lists below.
- **Geographic aggregation level** - National, Province/Territory (PT), CMA, Census Division (CD), or Census Subdivision (CSD).
- **Temporal aggregation level** - Monthly, Quarterly, or Annual.

Part 1 summarizes, for every (theme, series) pair in `series_facts` today, which breakdowns/geography/temporal grains it has. Part 2 works backwards from the source downloads (`data-raw/manifest.csv`, `data-raw/process_log.csv`, and the raw `catalog/facts` tables underneath `series_facts`) to check whether anything published upstream failed to reach the final database, and if so, why.

2 Part 1: Coverage by theme and series

`series_facts` currently holds 32 (theme, series) pairs across 7 themes. 6 of them have no `facet = "Total"` row at all (no member source table publishes an overall grand-total row for that series) - the app defaults "Break down by" to the first real breakdown instead of "None" for these, so they don't open on a blank chart (see `app/app.R`'s series/geo_level `observeEvent` handlers).

3 Part 2: Reconciling against the source downloads

3.1 2.1 Resources that never became a processed table

Of every resource in `data-raw/manifest.csv`, 4 did not become a processed long-format table:

`error` here is IRB's Data Dictionary resource (documentation, not a data table - its URL is dead on canada.ca; harmless, since its columns are already plain English). The `skipped_duplicate_sheet` rows are workbook sheets `R/process_utils.R` recognized as duplicating a richer sibling sheet in the same workbook during processing itself, before a `catalog` row was ever created for them.

Asylum Claimants

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
All claims	Age; Claim Office Type; Gender	National, PT	Monthly, Quarterly, Annual	2015–2026	3	yes
All claims (last known address)	(none)	National, PT	Annual	2015–2025	1	yes
Top 25 ranking by citizenship (annual)	Country of Citizenship	National, PT	Monthly, Quarterly, Annual	2015–2026	1	no
Top 25 ranking by citizenship (claim year)	Country of Citizenship	National	Monthly, Quarterly, Annual	2015–2026	1	no
Top 25 ranking by citizenship (monthly)	Country of Citizenship	National, PT	Monthly, Quarterly, Annual	2015–2026	2	no
Top 25 ranking by citizenship (most recent complete month)	Top 25 Countries of Citizenship in the Most Recent Month	National, PT	Monthly, Quarterly, Annual	2015–2026	2	yes
Top 25 ranking by citizenship (most recent complete year)	Top 25 Countries of Citizenship in the Most Recent Complete Year	National, PT	Monthly, Quarterly, Annual	2015–2026	2	yes

International Mobility Program

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
All permits	Country of Citizenship	National	Annual	2000–2024	1	yes
All permits (Year became effective)	Country of Citizenship; Gender; Intended Occupation (4-Digit NOC 2011); Program Level 2	National, PT	Monthly, Quarterly, Annual	2015–2026	4	yes
All permits (at December 31)	Program	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes
Permits for humanitarian and compassionate purposes (Year became effective)	Country of Citizenship; Gender	National	Monthly, Quarterly, Annual	2015–2026	2	yes
Permits for work purposes only	Country of Citizenship	National	Annual	2000–2024	1	yes
Permits for work purposes only (at December 31)	Program	National, PT	Annual	2000–2024	1	yes
Permits under post-grad employment	(none)	National, PT, CD	Monthly, Quarterly, Annual	2015–2026	1	yes
Permits under post-grad employment (Year became effective)	(none)	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes
Permits under post-grad employment (at December 31)	(none)	National, PT	Annual	2000–2024	1	yes

Permanent Residents

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
Adoptions via PR	Gender	National	Monthly, Quarterly, Annual	2015–2026	1	yes
All admissions	Age Group; Country of Citizenship; Gender; Immigration Category; Intended Occupation (NOC 2011)	National, PT, CD	Monthly, Quarterly, Annual	2015–2026	6	yes
Anglophone admissions within Québec	Immigration Category	National, PT	Monthly, Quarterly, Annual	2026–2026	2	yes
Francophone admissions outside Québec	Age Group; Country of Citizenship; Gender; Immigration Category	National, PT	Monthly, Quarterly, Annual	2015–2026	8	yes

Refugees

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
Country of Persecution	Country of Persecution	National	Annual	2013–2026	1	no
Outcome	Outcome	National	Annual	1997–2026	1	no
Province of Claimant	Province of Residence at the Time of the Refugee Claim Decision	National	Annual	1997–2021	1	no

Study Permit Holders

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
All permits (Year became effective)	Country of Citizenship; Gender; Study Level	National, PT	Monthly, Quarterly, Annual	2015–2026	3	yes
All permits (at December 31)	Country of Citizenship; Study Level	National, PT	Monthly, Quarterly, Annual	2000–2024	5	yes
Post-Secondary (Year became effective)	Citizenship; DLI Name	National, PT	Monthly, Quarterly, Annual	2016–2026	3	yes

Temporary Foreign Worker Program

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
All permits (Year became effective)	Country of Citizenship; Gender; Intended Occupation (4-Digit NOC 2011); Program	National, PT	Monthly, Quarterly, Annual	2015–2026	4	yes
All permits (at December 31)	Country of Citizenship; Program	National, PT	Annual	2000–2024	2	yes

TR → PR Transition

Series	Breakdowns	Geography	Temporal	Years	# tables	Total
Prior IMP, under post-grad	Immigration Category	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes
Prior IMP, work purposes only	Immigration Category	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes
Prior TFWP	Immigration Category	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes
Prior study permit	Immigration Category	National, PT	Monthly, Quarterly, Annual	2015–2026	1	yes

Resources excluded before processing

Dataset	Table (from manifest)	Status	Note
permanent_residents	Canada ex-QC - French-Speaking Permanent Residents by Province/Territory and Census Metropolitan Area	skipped_duplicate_sheet	
permanent_residents	Canada ex-QC - French-Speaking Permanent Residents by Province/Territory and Gender	skipped_duplicate_sheet	
permanent_residents	Canada ex-QC - French-Speaking Permanent Residents by Province/Territory and Immigration Category	skipped_duplicate_sheet	
rpd_decisions	Refugee Protection Division (RPD) Decisions - Data Dictionary	error	downloaded file not found

3.2 2.2 Tables intentionally excluded as redundant duplicates

A further 14 tables are processed successfully but then dropped by name in `data-raw/05_build_duckdb.R`'s `EXCLUDED_DUPLICATE_TABLE_IDS` list - each one verified (per that script's own comments) to be an exact or near-exact marginal of a richer sibling table already kept, adding no dimension the kept table lacks. These are a deliberate, documented dedup decision, not data loss - see that script for the specific value comparison behind each entry.

3.3 2.3 Tables that reached `catalog/facts` but contributed zero rows to `series_facts`

5 tables that made it as far as `catalog/facts` never supplied a single row to `series_facts` - every cell they publish was, for every period checked, matched or exceeded by some other member table of the same series:

None of these are silent data loss at the *series* level - in every case checked, the series' own Total facet is still correctly populated by another member table. The one exception, flagged inline above if present, is a genuine anomaly rather than a benign tie.

3.4 2.4 Breakdowns hidden behind generic level labels - found, and fixed

`guess_level_labels()` (`R/duckdb_utils.R`) derives each level's name by splitting a source table's title on its "by …" phrase - e.g. "by Country of Citizenship and Immigration Category" → two names. Where a table's row hierarchy actually nests *deeper* than the title spells out - a title that names a dimension once (e.g. "… and Program") when the data itself has that same dimension split across several nested sub-levels (Program → sub-program-type → specific program) - every level past what the title names falls back to a generic placeholder ("Category Level 3", "Program Level 4", …).

That placeholder text is also, separately, what used to mark a level as *filler* for the Break down by / Series Facts pipeline (`FILLER_LEVEL_REGEX` in `app/R/app_utils.R`) - deliberately, so that genuinely meaningless padding columns don't show up as a real breakdown choice. The two cases were indistinguishable from title text alone, so a table whose title merely under-named a real, deeply-nested dimension had that dimension's data **silently dropped** from `series_facts` entirely.

This is now fixed. `table_shape()` (`data-raw/08_build_series_facts.R`) no longer trusts a generic label's filler classification at face value: a new `level_has_real_data()` (`app/R/app_utils.R`) checks the level column's *actual content* in `facts` first, and only treats it as real filler when the column is genuinely empty (structural padding past a table's real depth). A generically-labeled column with real, populated values is now kept as a real facet position instead - still under its generic label (no title text exists to rename it from), but no longer discarded. Levels recovered this way are still subject to `MAX_BREAKDOWN_LEVELS`

Tables present in the raw layer but absent from series_facts

Theme	Series	Table	Reason
Permanent Residents	All admissions	canada permanent residents by pro…	Its own grand total exactly matches (or falls short of) another member table's for every year - never the least-suppressed sibling, so "largest value wins" never credits it. Not data loss: the series' Total facet is still correctly populated by the winning sibling.
Permanent Residents	Francophone admissions outside Québec	canada ex qc admissions of french…	Publishes no subtotal/grand-total row at any geographic depth (every row is a finest-grain leaf) - structurally cannot win the Total-facet merge, which only ever compares published subtotal rows.
Permanent Residents	Francophone admissions outside Québec	canada ex qc french speaking perm…	Its own grand total exactly matches (or falls short of) another member table's for every year - never the least-suppressed sibling, so "largest value wins" never credits it. Not data loss: the series' Total facet is still correctly populated by the winning sibling.
Permanent Residents	Francophone admissions outside Québec	canada ex qc percentage of french…	Publishes a computed *percentage* metric, not a count - structurally can't be compared to count-based siblings under "largest value wins", so it never contributes and never should.
International Mobility Program	Permits for humanitarian and compassionate purposes (Year became effective)	canada work permit holders for hu…	Its own grand total exactly matches (or falls short of) another member table's for every year - never the least-suppressed sibling, so "largest value wins" never credits it. Not data loss: the series' Total facet is still correctly populated by the winning sibling.

capping like any other facet (§2.5) - `permanent_residents/0f1e6cce` now logs exactly that capping warning, since its recovered breakdown reaches 4 levels.

12 tables were affected; 11 of them now contribute their previously-hidden breakdown to `series_facts` (verified directly against the rebuilt table, not just re-inferred from labels):

Two concrete examples, confirmed directly against the rebuilt `series_facts`:

- `permanent_residents/0f1e6cce` ("… by Country of Citizenship and Immigration Category") is 4 levels deep. Levels 1-2 (Country of Citizenship, Immigration Category) are named correctly; levels 3-4 fall back to "Category Level 3/4" - and hold real, populated values (`Provincial Nominee Program`, `Government-Assisted Refugee`, `Self-Employed`, …): a genuine Immigration-Category sub-program breakdown. It now shows up in `series_facts` as `breakdown_2`/`breakdown_3` under the `Country of Citizenship` facet (capped at `MAX_BREAKDOWN_LEVELS`, see §2.5 - this table logs that capping warning).
- `temporary_workers/520ac8e1` ("… by province/territory of intended destination, program and year in which permit(s) became effective") is 5 levels deep. Level 1 (Province/Territory) is geography; levels 2-5 all fall back to generic labels, but hold real Program category values (`Canadian Interests`, `Agreements`, `Other IMP Participants`, and their own sub-categories like `Significant benefit`, `Canada-International`). This table has no published grand-total row (see §2.3), so it still never contributes to the series' Total facet - but its Program breakdown now surfaces as a `Program Level 2` facet in its own right, where it was entirely absent before.

This was a title-parsing limitation, not a per-table one-off - any future table whose title names a dimension once but nests it several levels deep will be caught by the same content check now, rather than needing another one-off fix.

3.5 2.5 Breakdown-level capping (`MAX_BREAKDOWN_LEVELS`)

Separately from §2.4, `data-raw/08_build_series_facts.R` caps how many *correctly-named* nested breakdown levels it keeps per facet at `MAX_BREAKDOWN_LEVELS = 3` (`breakdown_1`/`breakdown_2`/`breakdown_3`) - anything deeper is summed away into `breakdown_3` rather than dropped, and the script logs a warning (`{table_id}: facet '...' has N levels, capping at 3`) whenever this actually happens. No such warning was logged in the build this report was generated against - every table's *correctly-labeled* breakdown currently fits within the cap. This is a distinct, narrower risk from §2.4: capping only ever applies to levels the pipeline recognizes as real breakdowns in the first place, and folds the excess into a coarser category rather than discarding it outright.

Theme	Table	Table title	Total levels	Generic levels	Recovered
Permanent Residents	canada ex qc percentage of french speakin…	Canada ex-QC - Percentage of French Speaking Permanent Residents by Immigration Category	3	2	no
Permanent Residents	canada permanent residents by country of …	Canada - Permanent Residents by Country of Citizenship and Immigration Category	4	2	yes
Permanent Residents	canada permanent residents by province te…	Canada - Permanent Residents by Province/Territory and Immigration Category	4	2	yes
Permanent Residents	canada ex qc french speaking permanent re…	Canada ex-QC - French-Speaking Permanent Residents by Province/Territory and Immigration Category — Pr Pt Immcat	4	2	yes
Permanent Residents	canada ex qc french speaking permanent re…	Canada ex-QC - French-Speaking Permanent Residents by Immigration Category	3	2	yes
TR → PR Transition	canada permanent residents with prior int…	Canada – Permanent Residents with Prior International Mobility Program (IMP) Work Permit Holder Status under Post-Graduate Employment by Province/Territory and Immigration Category	4	2	yes

Theme	Table	Table title	Total levels	Generic levels	Recovered
TR → PR Transition	canada permanent residents with prior int…	Canada – Permanent Residents with Prior International Mobility Program Work Permit Holder Status (for Work Purpose Only) by Province/Territory and Immigration Category	4	2	yes
TR → PR Transition	canada permanent residents with prior tem…	Canada – Permanent Residents with Prior Temporary Foreign Worker Program Work Permit Holder Status (for Work Purpose Only) by Province/Territory and Immigration Category	4	2	yes
TR → PR Transition	canada permanent residents with prior stu…	Canada – Permanent Residents with Prior Study Permit Holder Status by Province/Territory and Immigration Category	4	2	yes
temporary_workers	canada international mobility program wor…	Canada - International Mobility Program work permit holders by province/territory of intended destination, program and year in which permit(s) became effective	5	4	yes
temporary_workers	canada international mobility program wor…	Canada - International Mobility Program work permit holders (work purpose only) on December 31st by province/territory and program	4	2	yes
temporary_workers	canada international mobility program wor…	Canada - International Mobility Program work permit holders by province/territory, program and year in which permit(s) became effective	5	2	yes

4 Conclusion

One real breakdown-data-loss gap was found, and it's now fixed. The report above traces every way a source table can fail to reach `series_facts`:

- 4 source resources never became a processed table at all (§2.1) - one non-data documentation resource, plus 3 sheets a richer sibling in the same workbook already supersedes.
- 14 further tables are deliberately excluded post-processing as verified duplicates (§2.2) - documented, intentional, not a gap.
- 5 tables reach the raw `catalog/facts` layer but contribute zero rows to `series_facts` (§2.3) - in every case but any flagged anomaly, this is because another member table of the same series always matches or exceeds their published totals, so the series' own Total facet is unaffected.
- **12 tables had a real, populated nested breakdown that was invisible in `series_facts`** (§2.4), because `guess_level_labels()`'s title parsing under-names how deep the table's row hierarchy actually goes, and the resulting generic placeholder text was indistinguishable from genuine filler. This was the one finding here that constituted real breakdown data loss - `table_shape()` now checks actual column content (`level_has_real_data()`) before trusting a generic label's filler classification, and 11 of the 12 affected tables now contribute their previously-hidden breakdown (the remainder still don't contribute for the separate, unrelated reasons in §2.3 - e.g. no published grand-total row - not because of this fix).
- No table currently triggers `MAX_BREAKDOWN_LEVELS` capping beyond what's noted in §2.4 (§2.5).